MySQL Practice Sheet: INNER JOIN

1. Concept Overview

An INNER JOIN in MySQL is used to combine rows from two or more tables based on a related column between them. It returns only the records that have matching values in both tables.

Syntax:

```
SELECT table1.column_name, table2.column_name, ...
FROM table1
INNER JOIN table2
ON table1.common_column = table2.common_column;
```

2. Sample Tables

Table: Employees

EmpID	EmpName	DeptID
1	Amit Sharma	D101
2	Rina Kapoor	D102
3	Ravi Kumar	D103
4	Anita Verma	D104
5	Suresh Gupta	D105

Table: Departments

DeptID	DeptName	Location
D101	HR	Delhi
D102	IT	Mumbai
D103	Finance	Pune
D106	Admin	Chennai

3. Practice Queries

Q1. Display employee names along with their department names.

👉 SELECT Employees.EmpName, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;

Q2. Display employee names, department names, and locations.

👉 SELECT EmpName, DeptName, Location
FROM Employees INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;

Q3. Display names of employees working in the 'IT' department.

👉 SELECT EmpName
FROM Employees INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Departments.DeptName = 'IT';

Q4. Display names of employees and their department locations in 'Delhi'.

👉 SELECT EmpName, Location
FROM Employees INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Departments.Location = 'Delhi';

Q5. List employees and their department details for only matching DeptIDs.

👉 SELECT *
FROM Employees INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;

4. Task

- Create both tables in MySQL and insert the given records.
- Write queries using INNER JOIN to answer the above questions.
- Add 3 more queries of your own combining columns from both tables.
- Observe what happens when a DeptID does not match (e.g., D105, D106).